

Index	Monitoring value	Unit	Scale	ID	Description
V2.10.12	Pump 6 RunningTime	h	0.1	1625	
V2.10.13	Pump 7 RunningTime	h	0.1	1626	
V2.10.14	Pump 8 RunningTime	h	0.1	1627	

5.12 Monitoring of Maintenance Counters

Table 22: Maintenance Counter Monitoring

Index	Monitoring value	Unit	Scale	ID	Description
V2.11.1	Maintenance Counter 1	h/kRev	Varies	1101	

5.13 Fieldbus Process Data Monitoring

Table 23: Fieldbus Process Data Monitoring

Index	Monitoring value	Unit	Scale	ID	Description
V2.12.1	FB Control Word			874	
V2.12.2	FB Status Word			864	
V2.12.3	FB Speed Reference			875	
V2.12.4	FB Speed Actual			865	
V2.12.5	Fieldbus General Control Word			861	
V2.12.6	Fieldbus General Status Word			862	

Table 24: Fieldbus Data In

Index	Monitoring value	Unit	Scale	ID	Description
V2.12.7.1	FB Data In 1		1	876	
V2.12.7.2	FB Data In 2		1	877	
V2.12.7.3	FB Data In 3		1	878	
V2.12.7.4	FB Data In 4		1	879	
V2.12.7.5	FB Data In 5		1	880	
V2.12.7.6	FB Data In 6		1	881	
V2.12.7.7	FB Data In 7		1	882	
V2.12.7.8	FB Data In 8		1	883	
V2.12.7.9	FB Data In 9		1	229	
V2.12.7.10	FB Data In 10		1	230	
V2.12.7.11	FB Data In 11		1	231	
V2.12.7.12	FB Data In 12		1	232	
V2.12.7.13	FB Data In 13		1	233	

Index	Monitoring value	Unit	Scale	ID	Description
V2.12.7.14	FB Data In 14		1	234	
V2.12.7.15	FB Data In 15		1	235	
V2.12.7.16	FB Data In 16		1	236	

Table 25: Fieldbus Data Out

Index	Monitoring value	Unit	Scale	ID	Description
V2.12.8.1	FB Data Out 1		1	866	
V2.12.8.2	FB Data Out 2		1	867	
V2.12.8.3	FB Data Out 3		1	868	
V2.12.8.4	FB Data Out 4		1	869	
V2.12.8.5	FB Data Out 5		1	870	
V2.12.8.6	FB Data Out 6		1	871	
V2.12.8.7	FB Data Out 7		1	872	
V2.12.8.8	FB Data Out 8		1	873	
V2.12.8.9	FB Data Out 9		1	245	
V2.12.8.10	FB Data Out 10		1	246	
V2.12.8.11	FB Data Out 11		1	247	
V2.12.8.12	FB Data Out 12		1	248	
V2.12.8.13	FB Data Out 13		1	249	
V2.12.8.14	FB Data Out 14		1	250	
V2.12.8.15	FB Data Out 15		1	251	
V2.12.8.16	FB Data Out 16		1	252	

5.14 Drive Customizer Monitoring

Table 26: Drive Customizer Monitoring

Index	Monitoring value	Unit	Scale	ID	Description
V2.13.2	Block Out.1			15020	
V2.13.3	Block Out.2			15040	
V2.13.4	Block Out.3			15060	
V2.13.5	Block Out.4			15080	
V2.13.6	Block Out.5			15100	
V2.13.7	Block Out.6			15120	
V2.13.8	Block Out.7			15140	

Index	Monitoring value	Unit	Scale	ID	Description
V2.13.9	Block Out.8			15160	
V2.13.10	Block Out.9			15180	
V2.13.11	Block Out.10			15200	

5.15 Condition-based Monitoring

Table 27: Condition-based Monitoring

Index	Name	Min	Max	Unit	Default	ID	Description
V2.16.1 Baseline Data							
V2.16.1.1	BaselineDataElem1	-300.00	320.00	various	0.00	3601	
V2.16.1.2	BaselineDataElem2	-300.00	320.00	various	0.00	3602	
V2.16.1.3	BaselineDataElem3	-300.00	320.00	various	0.00	3603	
V2.16.1.4	BaselineDataElem4	-300.00	320.00	various	0.00	3604	
V2.16.1.5	BaselineDataElem5	-300.00	320.00	various	0.00	3605	
V2.16.1.6	BaselineDataElem6	-300.00	320.00	various	0.00	3606	
V2.16.1.7	BaselineDataElem7	-300.00	320.00	various	0.00	3607	
V2.16.1.8	BaselineDataElem8	-300.00	320.00	various	0.00	3608	
V2.16.1.9	BaselineDataElem9	-300.00	320.00	various	0.00	3609	
V2.16.1.10	BaselineDataElem10	-300.00	320.00	various	0.00	3610	
V2.16.1.11	BaselineRun Status	0	14		0	3622	0=Not Started 1=Running 2=Running 10% 3=Running 20% 4=Running 30% 5=Running 40% 6=Running 50% 7=Running 60% 8=Running 70% 9=Running 80% 10=Running 90% 11=Completed 12=Run Failed 13=Manual Before Baseline 14=Manual After Baseline
V2.16.2 Stator Winding Monitor							
V2.16.2.1	Current Unbalance	0.00	100.00	%	0.00	3617	
V2.16.2.2	Current Threshold Value	0.00	100.00	%	0.00	3620	
V2.16.2.3	Current Warning S1 High Threshold	0.00	100.00	%	0.00	3611	
V2.16.2.4	Current Warning S2 High Threshold	0.00	100.00	%	0.00	3612	

Index	Name	Min	Max	Unit	Default	ID	Description
V2.16.2.5	Current Alarm High Threshold	0.00	100.00	%	0.00	3613	
V2.16.3 Vibration							
V2.16.3.1	Vibration	0.00	100.00	%	0.00	3623	
V2.16.3.2	Vibration Threshold Value	0.00	100.00	%	0.00	3624	
V2.16.3.3	Vibration Warning S1 High Threshold	0.00	100.00	%	0.00	3625	
V2.16.3.4	Vibration Warning S2 High Threshold	0.00	100.00	%	0.00	3626	
V2.16.3.5	Vibration Alarm High Threshold	0.00	100.00	%	0.00	3627	
V2.16.4 Load							
V2.16.4.1	Motor Torque	-300.0	300.0	%	0.0	4	
V2.16.4.2	Load Threshold Value	-300.0	300.0	%	0.00	3628	
V2.16.4.3	Load Warning S1 High Threshold	-300.0	300.0	%	0.00	3629	
V2.16.4.4	Load Warning S2 High Threshold	-300.0	300.0	%	0.00	3630	
V2.16.4.5	Load Alarm High Threshold	-300.0	300.0	%	0.00	3631	
V2.16.4.6	Load Warning S1 Low Threshold	-300.0	300.0	%	0.00	3632	
V2.16.4.7	Load Warning S2 Low Threshold	-300.0	300.0	%	0.00	3633	
V2.16.4.8	Load Alarm Low Threshold	-300.0	300.0	%	0.00	3634	
V2.16.5	Condition Based Monitoring Status	0	65535		0	3619	B0= Stator current warning S1 B2= Vibration warning S1 B3= Load warning S1 B5= Stator current warning S2 B7= Vibration warning S2 B8= Load warning S2 B10= Stator current alarm/fault B12= Vibration alarm/fault B13= Load alarm/fault

6 Parameters Menu

6.1 Group 3.1: Motor Settings

Table 28: Motor Nameplate Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.1.1.1	Motor Nominal Voltage	Varies	Varies	V	Varies	110	
P3.1.1.2	Motor Nominal Frequency	8.00	320.00	Hz	50 / 60	111	
P3.1.1.3	Motor Nominal Speed	24	19200	rpm	Varies	112	
P3.1.1.4	Motor Nominal Current	IH * 0.1	IH * 2	A	Varies	113	
P3.1.1.5	Motor Cos Phi (Power Factor)	0.30	1.00		Varies	120	
P3.1.1.6	Motor Nominal Power	Varies	Varies	kW	Varies	116	

Table 29: Motor Control Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.1.2.2	Motor Type	0	2		0	650	0 = Induction motor 1 = PM motor 2 = Reluctance motor
P3.1.2.3	Switching Frequency	1.5	Varies	kHz	Varies	601	
P3.1.2.4	Identification	0	2		0	631	0 = No action 1 = At standstill 2 = With rotation
P3.1.2.5	Magnetizing Current	0.0	2*IH	A	0.0	612	
P3.1.2.6	Motor Switch	0	1		0	653	0 = Disabled 1 = Enabled
P3.1.2.10	Overvoltage Control	0	1		1	607	0 = Disabled 1 = Enabled
P3.1.2.11	Undervoltage Control	0	1		1	608	0 = Disabled 1 = Enabled
P3.1.2.12	Energy Optimization	0	1		0	666	0 = Disabled 1 = Enabled
P3.1.2.13	Stator Voltage Adjust	50.0	150.0	%	100.0	659	
P3.1.2.15	Rs Voltage Drop	0.00	50.00	%	0.00	662	
P3.1.2.16	Ls Voltage Drop	0.00	200.00	%	0.00	673	
P3.1.2.17	Motor BEM Voltage	0	Varies	V	0	674	

Table 30: Motor Limit Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.1.3.1	Motor Current Limit	IH*0.1	IS	A	Varies	107	
P3.1.3.2	Motor Torque Limit	0.0	300.0	%	300.0	1287	

Table 31: Open Loop Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.1.4.1	U/f Ratio	0	2		0	108	0 = Linear 1 = Squared 2 = Programmable
P3.1.4.2	Field Weakening Point Frequency	8.00	P3.3.1.2	Hz	Varies	602	
P3.1.4.3	Voltage at Field Weakening Point	10.00	200.00	%	100.00	603	
P3.1.4.4	U/f Midpoint Frequency	0.00	P3.1.4.2	Hz	Varies	604	
P3.1.4.5	U/f Midpoint Voltage	0.0	100.0	%	100.00	605	
P3.1.4.6	Zero Frequency Voltage	0.00	40.00	%	Varies	606	
P3.1.4.7	Flying Start Options	0	255		0	1590	B0 = Search the shaft frequency only from the same direction as the frequency reference B1 = Disable AC scanning B4 = Use the frequency reference for the initial guess B5 = Disable DC pulses B6 = Flux build with current control B7 = Reverse injection direction
P3.1.4.8	Flying Start Scan Current	0.0	100.0	%	Varies	1610	
P3.1.4.9	Start Boost	0	1		0	109	0 = Disabled 1 = Enabled
M3.1.4.12	I/f Start	This menu includes 3 parameters. See Table 32 .					

Table 32: I/f Start Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.1.4.12.1	I/f Start	0	1		0	534	0 = Disabled 1 = Enabled
P3.1.4.12.2	I/f Start Frequency	5.0	0.5 * P3.1.1.2		0.2 * P3.1.1.2	535	
P3.1.4.12.3	I/f Start Current	0.0	150.0	%	80.0	536	

Table 33: Torque Stabilator Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.1.4.13.1	Torque Stabilator Gain	0.0	500.0	%	50.0	1412	
P3.1.4.13.2	Torque Stabilator Gain at Field Weakening Point	0.0	500.0	%	50.0	1414	
P3.1.4.13.3	Torque Stabilator Damping Time Constant	0.0005	1.0000	s	0.0050	1413	
P3.1.4.13.4	Torque Stabilator Damping Time Constant for PM motors	0.0005	1.0000	s	0.0500	1735	
P3.1.4.13.5	Torque Stabilator Filter Time Constant	0.0000	0.5000	s	0.0005	15650	
P3.1.4.13.6	Torque Stabilator Filter Time Constant for PM motors	0.0000	0.5000	s	0.0005	15651	

6.2 Group 3.2: Start/Stop Setup

Table 34: Start/Stop Setup

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.2.1	Remote Control Place	0	1		0 ⁽¹⁾	172	0 = I/O control 1 = Fieldbus control
P3.2.2	Local/Remote	0	1		0 ⁽¹⁾	211	0 = Remote 1 = Local
P3.2.3	Keypad Stop Button	0	1		0	114	0 = Yes 1 = No
P3.2.4	Start Function	0	1		0	505	0 = Ramping 1 = Flying start
P3.2.5	Stop Function	0	1		0	506	0 = Coasting 1 = Ramping
P3.2.6	I/O A Start/Stop Logic	0	4		2 ⁽¹⁾	300	Logic = 0 - Ctrl sgn 1 = Forward - Ctrl sgn 2 = Backward Logic = 1 - Ctrl sgn 1 = Forward (edge) - Ctrl sgn 2 = Inverted Stop - Ctrl sgn 3 = Bckwrđ (edge) Logic = 2 - Ctrl sgn 1 = Forward (edge) - Ctrl sgn 2 = Bckwrđ (edge) Logic = 3 - Ctrl sgn 1 = Start - Ctrl sgn 2 = Reverse Logic = 4

Index	Parameter	Min	Max	Unit	Default	ID	Description
							- Ctrl sgn 1 = Start (edge) - Ctrl sgn 2 = Reverse
P3.2.7	I/O B Start/Stop Logic	0	4		2 ⁽¹⁾	363	See P3.2.6.
P3.2.8	Fieldbus Start Logic	0	1		0	889	0 = A rising edge is necessary 1 = State
P3.2.9	Start Delay	0.000	60.000	s	0.000	524	
P3.2.10	Remote to Local Function	0	2		2	181	0 = Keep Run 1 = Keep Run & Reference 2 = Stop
P3.2.11	Restart Delay	0.0	20.0	min	0.0	15555	0 = Not used

¹ The selection of the application with parameter P1.2 Application gives the default value. See [11.25 Default Values of Parameters in the Different Applications](#).

6.3 Group 3.3: References

Table 35: Frequency Reference Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.3.1.1	Minimum Frequency Reference	0.00	P3.3.1.2	Hz	0.00	101	
P3.3.1.2	Maximum Frequency Reference	P3.3.1.1	320.00	Hz	50.00/60.00	102	
P3.3.1.3	Positive Frequency Reference Limit	-320.0	320.0	Hz	320.00	1285	
P3.3.1.4	Negative Frequency Reference Limit	-320.0	320.0	Hz	-320.00	1286	
P3.3.1.5	I/O Control Reference A Selection	1	20		6 ⁽¹⁾	117	1 = Preset Frequency 0 2 = Keypad reference 3 = Fieldbus 4 = AI1 5 = AI2 6 = AI1+AI2 7 = PID reference 8 = Motor potentiometer 11 = Block Out.1 12 = Block Out.2 13 = Block Out.3 14 = Block Out.4 15 = Block Out.5 16 = Block Out.6 17 = Block Out.7 18 = Block Out.8 19 = Block Out.9 20 = Block Out.10

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.3.1.6	I/O Control Reference B Selection	1	20		4 ⁽¹⁾	131	See P3.3.1.5.
P3.3.1.7	Keypad Control Reference Selection	1	20		2 ⁽¹⁾	121	1 = Preset Frequency 0 2 = Keypad reference 3 = Fieldbus 4 = AI1 5 = AI2 6 = AI1+AI2 7 = PID reference 8 = Motor potentiometer 11 = Block Out.1 12 = Block Out.2 13 = Block Out.3 14 = Block Out.4 15 = Block Out.5 16 = Block Out.6 17 = Block Out.7 18 = Block Out.8 19 = Block Out.9 20 = Block Out.10
P3.3.1.8	Keypad Reference	0.00	P3.3.1.2	Hz	0.00	184	
P3.3.1.9	Keypad Direction	0	1		0	123	0 = Forward 1 = Reverse
P3.3.1.10	Fieldbus Control Reference Selection	1	20		3 ⁽¹⁾	122	1 = Preset Frequency 0 2 = Keypad reference 3 = Fieldbus 4 = AI1 5 = AI2 6 = AI1+AI2 7 = PID reference 8 = Motor potentiometer 11 = Block Out.1 12 = Block Out.2 13 = Block Out.3 14 = Block Out.4 15 = Block Out.5 16 = Block Out.6 17 = Block Out.7 18 = Block Out.8 19 = Block Out.9 20 = Block Out.10

¹ The selection of the application with parameter P1.2 Application gives the default value. See the default values in [11.25 Default Values of Parameters in the Different Applications](#).

Table 36: Preset Frequency Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.3.3.1	Preset Frequency Mode	0	1		0 ⁽¹⁾	182	0 = Binary coded 1 = Number of inputs
P3.3.3.2	Preset Frequency 0	P3.3.1.1	P3.3.1.2	Hz	5.00	180	
P3.3.3.3	Preset Frequency 1	P3.3.1.1	P3.3.1.2	Hz	10.00 ⁽¹⁾	105	
P3.3.3.4	Preset Frequency 2	P3.3.1.1	P3.3.1.2	Hz	15.00 ⁽¹⁾	106	
P3.3.3.5	Preset Frequency 3	P3.3.1.1	P3.3.1.2	Hz	20.00 ⁽¹⁾	126	
P3.3.3.6	Preset Frequency 4	P3.3.1.1	P3.3.1.2	Hz	25.00 ⁽¹⁾	127	
P3.3.3.7	Preset Frequency 5	P3.3.1.1	P3.3.1.2	Hz	30.00 ⁽¹⁾	128	
P3.3.3.8	Preset Frequency 6	P3.3.1.1	P3.3.1.2	Hz	40.00 ⁽¹⁾	129	
P3.3.3.9	Preset Frequency 7	P3.3.1.1	P3.3.1.2	Hz	50.00 ⁽¹⁾	130	
P3.3.3.10	Preset Frequency Selection 0				DigIN SlotA.4	419	
P3.3.3.11	Preset Frequency Selection 1				DigIN SlotA.5	420	
P3.3.3.12	Preset Frequency Selection 2				DigIN Slot0.1	421	

¹ The selection of the application with parameter P1.2 Application gives the default value. See the default values in [11.25 Default Values of Parameters in the Different Applications](#).

Table 37: Motor Potentiometer Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.3.4.1	Motor Potentiometer UP				DigIN Slot0.1	418	OPEN = Not active CLOSED = Active
P3.3.4.2	Motor Potentiometer DOWN				DigIN Slot0.1	417	OPEN = Not active CLOSED = Active
P3.3.4.3	Motor Potentiometer Ramp Time	0.1	500.0	Hz/s	10.0	331	
P3.3.4.4	Motor Potentiometer Reset	0	2		1	367	0 = No reset 1 = Reset if stopped 2 = Reset if powered down

Table 38: Flushing Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.3.6.1	Flushing ReferenceActivation				DigINSlot0.1 *	530	
P3.3.6.2	Flushing reference	-MaxRef	MaxRef	Hz	0.00 *	1239	

6.4 Group 3.4: Ramps and Brakes Setup

Table 39: Ramp 1 Setup

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.4.1.1	Ramp 1 Shape	0.0	100.0	%	0.0	500	
P3.4.1.2	Acceleration Time 1	0.1	3000.0	s	5.0	103	
P3.4.1.3	Deceleration Time 1	0.1	3000.0	s	5.0	104	

Table 40: Ramp 2 Setup

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.4.2.1	Ramp 2 Shape	0.0	100.0	%	0.0	501	
P3.4.2.2	Acceleration Time 2	0.1	3000.0	s	10.0	502	
P3.4.2.3	Deceleration Time 2	0.1	3000.0	s	10.0	503	
P3.4.2.4	Ramp 2 Selection	Varies	Varies		DigIN Slot0.1	408	OPEN = Ramp 1 Shape, Acceleration Time 1 and Deceleration Time 1. CLOSED = Ramp 2 Shape, Acceleration Time 2 and Deceleration Time 2.
P3.4.2.5	Ramp 2 Threshold Frequency	0.0	P3.3.1.2	Hz	0.0	533	0 = Not used

Table 41: Start Magnetization Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.4.3.1	Start Magnetizing Current	0.00	IL	A	IH	517	0 = Disabled
P3.4.3.2	Start Magnetizing Time	0.00	600.00	s	0.00	516	

Table 42: DC Brake Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.4.4.1	DC Brake Current	0	IL	A	IH	507	0 = Disabled
P3.4.4.2	DC Braking Time at Stop	0.00	600.00	s	0.00	508	0 = DC braking not used
P3.4.4.3	Frequency to Start DC Braking at Ramp Stop	0.10	10.00	Hz	1.50	515	

Table 43: Flux Braking Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.4.5.1	Flux Braking	0	1		0	520	0 = Disabled 1 = Enabled
P3.4.5.2	Flux Braking Current	0	IL	A	IH	519	

6.5 Group 3.5: I/O Configuration

Table 44: Digital Input Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.5.1.1	Control Signal 1 A				DigIN SlotA.1 ⁽¹⁾	403	
P3.5.1.2	Control Signal 2 A				DigIN SlotA.2 ⁽¹⁾	404	
P3.5.1.3	Control Signal 3 A				DigIN Slot0.1	434	
P3.5.1.4	Control Signal 1 B				DigIN Slot0.1 ⁽¹⁾	423	
P3.5.1.5	Control Signal 2 B				DigIN Slot0.1 ⁽¹⁾	424	
P3.5.1.6	Control Signal 3 B				DigIN Slot0.1	435	
P3.5.1.7	I/O B Control Force				DigIN Slot0.1 ⁽¹⁾	425	CLOSED = Force the control place to I/O B
P3.5.1.8	I/O B Reference Force				DigIN Slot0.1 ⁽¹⁾	343	CLOSED = I/O reference B (P3.3.1.6) gives the frequency reference
P3.5.1.9	Fieldbus Control Force				DigIN Slot0.1 ⁽¹⁾	411	
P3.5.1.10	Keypad Control Force				DigIN Slot0.1 ⁽¹⁾	410	
P3.5.1.11	External Fault Close				DigIN SlotA.3 ⁽¹⁾	405	OPEN = OK CLOSED = External fault
P3.5.1.12	External Fault Open				DigIN Slot0.2	406	OPEN = External fault CLOSED = OK
P3.5.1.13	Fault Reset Close				Varies	414	CLOSED = Resets all active faults
P3.5.1.14	Fault Reset Open				DigIN Slot0.1	213	OPEN = Resets all active faults
P3.5.1.15	Run Enable				DigIN Slot0.2	407	
P3.5.1.16	Run Interlock 1				DigIN Slot0.2	1041	
P3.5.1.17	Run Interlock 2				DigIN Slot0.2	1042	
P3.5.1.18	Motor Preheat ON				DigIN Slot0.1	1044	OPEN = No action CLOSED = Uses the DC current of the motor preheat in Stop state. Used when the value of P3.18.1 is 2.
P3.5.1.19	Ramp 2 Selection				DigIN Slot0.1	408	OPEN = Ramp 1 Shape, Acceleration Time 1, and Deceleration Time 1. CLOSED = Ramp 2 Shape, Acceleration Time 2, and Deceleration Time 2.
P3.5.1.20	Acc/Dec Prohibit				DigIN Slot0.1	415	

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.5.1.21	Preset Frequency Selection 0				DigIN SlotA.4 ⁽¹⁾	419	
P3.5.1.22	Preset Frequency Selection 1				Varies	420	
P3.5.1.23	Preset Frequency Selection 2				DigIN Slot0.1 ⁽¹⁾	421	
P3.5.1.24	Motor Potentiometer UP				DigIN Slot0.1	418	OPEN = Not active CLOSED = Active
P3.5.1.25	Motor Potentiometer DOWN				DigIN Slot0.1	417	OPEN = Not active CLOSED = Active
P3.5.1.26	Quick Stop Activation				Varies	1213	OPEN = Activated
P3.5.1.27	Timer 1				DigIN Slot0.1	447	
P3.5.1.28	Timer 2				DigIN Slot0.1	448	
P3.5.1.29	Timer 3				DigIN Slot0.1	449	
P3.5.1.30	PID Setpoint Boost				DigIN Slot0.1	1046	OPEN = No boost CLOSED = Boost
P3.5.1.31	PID Select Setpoint				DigIN Slot0.1 ⁽¹⁾	1047	OPEN = Setpoint 1 CLOSED = Setpoint 2
P3.5.1.32	External PID Start Signal				DigIN Slot0.2	1049	OPEN = PID2 in stop mode CLOSED = PID2 regulating
P3.5.1.33	External PID Select Setpoint				DigIN Slot0.1	1048	OPEN = Setpoint 1 CLOSED = Setpoint 2
P3.5.1.34	Reset Maintenance Counter 1				DigIN Slot0.1	490	CLOSED = Reset
P3.5.1.36	Flushing Reference Activation				DigIN Slot0.1 ⁽¹⁾	530	
P3.5.1.38	Fire Mode Activation OPEN				DigIN Slot0.2	1596	OPEN = Fire mode active CLOSED = No action
P3.5.1.39	Fire Mode Activation CLOSE				DigIN Slot0.1	1619	OPEN = No action CLOSED = Fire Mode active
P3.5.1.40	Fire Mode Reverse				DigIN Slot0.1	1618	OPEN = Forward CLOSED = Reverse
P3.5.1.41	Auto-cleaning Activation				DigIN Slot0.1	1715	
P3.5.1.42	Pump 1 Interlock				DigIN Slot0.1 ⁽¹⁾	426	OPEN = Not active CLOSED = Active
P3.5.1.43	Pump 2 Interlock				DigIN Slot0.1 ⁰	427	OPEN = Not active CLOSED = Active

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.5.1.44	Pump 3 Interlock				DigIN Slot0.1 ⁰	428	OPEN = Not active CLOSED = Active
P3.5.1.45	Pump 4 Interlock				DigIN Slot0.1	429	OPEN = Not active CLOSED = Active
P3.5.1.46	Pump 5 Interlock				DigIN Slot0.1	430	OPEN = Not active CLOSED = Active
P3.5.1.47	Pump 6 Interlock				DigIN Slot0.1	486	OPEN = Not active CLOSED = Active
P3.5.1.48	Pump 7 Interlock				DigIN Slot0.1	487	OPEN = Not active CLOSED = Active
P3.5.1.49	Pump 8 Interlock				DigIN Slot0.1	488	OPEN = Not active CLOSED = Active
P3.5.1.52	Reset kWh Trip Counter				DigIN Slot0.1	1053	
P3.5.1.53	Parameter Set 1/2 Selection				DigIN Slot0.1	496	OPEN = Parameter Set 1 CLOSED = Parameter Set 2
P3.5.1.54	Multi-Setpoint Selection 0				DigIN Slot0.1	15576	
P3.5.1.55	Multi-Setpoint Selection 1				DigIN Slot0.1	15577	
P3.5.1.56	Multi-Setpoint Selection 2				DigIN Slot0.1	15578	
P3.5.1.57	Multi-Setpoint Selection 3				DigIN Slot0.1	15579	
P3.5.1.58	Flow Switch				DigIN Slot0.2	15590	
P3.5.1.59	AHF Over Temperature				DigIN Slot0.2	15513	

¹ The selection of the application with parameter P1.2 Application gives the default value. See the default values in [11.25 Default Values of Parameters in the Different Applications](#).

NOTICE

Your option board and board setup give the number of available analogue inputs. The standard I/O board has 2 analog inputs.

Table 45: Analog Input 1 Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.5.2.1.1	AI1 Signal Selection				AnIN SlotA.1 ⁽¹⁾	377	
P3.5.2.1.2	AI1 Signal Filter Time	0.00	300.00	s	0.1 ⁽¹⁾	378	
P3.5.2.1.3	AI1 Signal Range	0	1		0 ⁽¹⁾	379	0 = 0...10 V/0...20 mA 1 = 2...10 V/4...20 mA

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.5.2.1.4	AI1 Custom. Min	-160.00	160.00	%	0.00 ⁽¹⁾	380	
P3.5.2.1.5	AI1 Custom. Max	-160.00	160.00	%	100.00 ⁽¹⁾	381	
P3.5.2.1.6	AI1 Signal Inversion	0	1		0 ⁽¹⁾	387	0 = Normal 1 = Signal inverted

¹ The selection of the application with parameter P1.2 Application gives the default value. See the default values in [11.25 Default Values of Parameters in the Different Applications](#).

Table 46: Analog Input 2 Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.5.2.2.1	AI2 Signal Selection				AnIN SlotA.2 ⁽¹⁾	388	See P3.5.2.1.1.
P3.5.2.2.2	AI2 Signal Filter Time	0.00	300.00	s	0.1 ⁽¹⁾	389	See P3.5.2.1.2.
P3.5.2.2.3	AI2 Signal Range	0	1		1 ⁽¹⁾	390	See P3.5.2.1.3.
P3.5.2.2.4	AI2 Custom. Min	-160.00	160.00	%	0.00 ⁽¹⁾	391	See P3.5.2.1.4.
P3.5.2.2.5	AI2 Custom. Max	-160.00	160.00	%	100.00 ⁽¹⁾	392	See P3.5.2.1.5.
P3.5.2.2.6	AI2 Signal Inversion	0	1		0 ⁽¹⁾	398	See P3.5.2.1.6.

¹ The selection of the application with parameter P1.2 Application gives the default value. See the default values in [11.25 Default Values of Parameters in the Different Applications](#).

Table 47: Analog Input 3 Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.5.2.3.1	AI3 Signal Selection				AnIN SlotD.1	141	See P3.5.2.1.1.
P3.5.2.3.2	AI3 Signal Filter Time	0.00	300.00	s	0.1	142	See P3.5.2.1.2.
P3.5.2.3.3	AI3 Signal Range	0	1		0	143	See P3.5.2.1.3.
P3.5.2.3.4	AI3 Custom. Min	-160.00	160.00	%	0.00	144	See P3.5.2.1.4.
P3.5.2.3.5	AI3 Custom. Max	-160.00	160.00	%	100.00	145	See P3.5.2.1.5.
P3.5.2.3.6	AI3 Signal Inversion	0	1		0	151	See P3.5.2.1.6.

Table 48: Analog Input 4 Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.5.2.4.1	AI4 Signal Selection				AnIN SlotD.2	152	See P3.5.2.1.1.
P3.5.2.4.2	AI4 Signal Filter Time	0.00	300.00	s	0.1	153	See P3.5.2.1.2.

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.5.2.4.3	AI4 Signal Range	0	1		0	154	See P3.5.2.1.3.
P3.5.2.4.4	AI4 Custom. Min	-160.00	160.00	%	0.00	155	See P3.5.2.1.4.
P3.5.2.4.5	AI4 Custom. Max	-160.00	160.00	%	100.00	156	See P3.5.2.1.5.
P3.5.2.4.6	AI4 Signal Inversion	0	1		0	162	See P3.5.2.1.6.

Table 49: Analog Input 5 Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.5.2.5.1	AI5 Signal Selection				AnIN SlotE.1	188	See P3.5.2.1.1.
P3.5.2.5.2	AI5 Signal Filter Time	0.00	300.00	s	0.1	189	See P3.5.2.1.2.
P3.5.2.5.3	AI5 Signal Range	0	1		0	190	See P3.5.2.1.3.
P3.5.2.5.4	AI5 Custom. Min	-160.00	160.00	%	0.00	191	See P3.5.2.1.4.
P3.5.2.5.5	AI5 Custom. Max	-160.00	160.00	%	100.00	192	See P3.5.2.1.5.
P3.5.2.5.6	AI5 Signal Inversion	0	1		0	198	See P3.5.2.1.6.

Table 50: Analog Input 6 Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.5.2.6.1	AI6 Signal Selection				AnIN SlotE.2	199	See P3.5.2.1.1.
P3.5.2.6.2	AI6 Signal Filter Time	0.00	300.00	s	0.1	200	See P3.5.2.1.2.
P3.5.2.6.3	AI6 Signal Range	0	1		0	201	See P3.5.2.1.3.
P3.5.2.6.4	AI6 Custom. Min	-160.00	160.00	%	0.00	203	See P3.5.2.1.4.
P3.5.2.6.5	AI6 Custom. Max	-160.00	160.00	%	100.00	204	See P3.5.2.1.5.
P3.5.2.6.6	AI6 Signal Inversion	0	1		0	209	See P3.5.2.1.6.

Table 51: Digital Output Settings on Standard I/O Board, Slot B

Index	Parameter	Min	Max	Unit	De- fault	ID	Description
P3.5.3.2.1	RO1 Func- tion	0	75		Varies	11001	The function selection for R01: 0 = None 1 = Ready 2 = Run 3 = General fault 4 = General fault inverted 5 = General alarm 6 = Reversed

Index	Parameter	Min	Max	Unit	De- fault	ID	Description
							7 = At speed 8 = Thermistor fault 9 = Motor regulator active 10 = Start signal active 11 = Keypad control active 12 = I/O B control activated 13 = Limit supervision 1 14 = Limit supervision 2 15 = Fire Mode active 16 = Flushing activated 17 = Preset freq. active 18 = Quick stop activated 19 = PID in Sleep mode 20 = PID soft fill active 21 = PID feedback supervision (limits) 22 = Ext. PID supervision (limits) 23 = Input press. alarm/fault 24 = Frost prot. alarm/fault 25 = Time channel 1 26 = Time channel 2 27 = Time channel 3 28 = FB ControlWord B13 29 = FB ControlWord B14 30 = FB ControlWord B15 31 = FB Process- Data1.B0 32 = FB Process- Data1.B1 33 = FB Process- Data1.B2 34 = Maintenance alarm 35 = Maintenance fault 37 = Block 2 Out 38 = Block 3 Out 39 = Block 4 Out 40 = Block 5 Out 41 = Block 6 Out 42 = Block 7 Out 43 = Block 8 Out 44 = Block 9 Out 45 = Block 10 Out 46 = Jockey pump control 47 = Priming pump control 48 = Auto-cleaning active 49 = Multi-pump K1 control 50 = Multi-pump K2 control 51 = Multi-pump K3 control 52 = Multi-pump K4 control 53 = Multi-pump K5 control 54 = Multi-pump K6 control 55 = Multi-pump K7 control 56 = Multi-pump K8 control 69 = Selected parameter set

Index	Parameter	Min	Max	Unit	De- fault	ID	Description
							72 = AHF Cap Disconnect 73 = AHF Cap Disconnect Inv 74 = Run indication 75 = Motor Switch Open
P3.5.3.2.2	RO1 ON Delay	0.00	320.00	s	0.00	11002	
P3.5.3.2.3	RO1 OFF Delay	0.00	320.00	s	0.00	11003	
P3.5.3.2.4	RO2 Function	0	56		Varies	11004	See P3.5.3.2.1.
P3.5.3.2.5	RO2 ON Delay	0.00	320.00	s	0.00	11005	See P3.5.3.2.2.
P3.5.3.2.6	RO2 OFF Delay	0.00	320.00	s	0.00	11006	See P3.5.3.2.3.
P3.5.3.2.7	RO3 Function	0	56		Varies	11007	See P3.5.3.2.1. Not visible if only 2 output relays are installed.
P3.5.3.2.8	RO3 ON Delay	0.00	320.00	s	0.00	11008	See P3.5.3.2.2. Not visible if only 2 output relays are installed.
P3.5.3.2.9	RO3 OFF Delay	0.00	320.00	s	0.00	11009	See P3.5.3.2.3. Not visible if only 2 output relays are installed.

The Digital Outputs of the Expander Slots C, D, and E

NOTICE

Shows only the parameters for the outputs on option boards in slots C, D, and E. Make the selections as in Basic AO1 Function (P3.5.4.1.1). This group or these parameters are not visible if there are no digital outputs in slots C, D, or E.

Table 52: Analog Output Settings on Standard I/O board

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.5.4.1.1	AO1 function	0	31		2 ⁽¹⁾	10050	0 = TEST 0% (Not used) 1 = TEST 100% 2 = Output freq (0 - fmax) 3 = Freq reference (0 - fmax) 4 = Motor speed (0 - Motor nominal speed) 5 = Output current (0 - InMotor) 6 = Motor torque (0 - TnMotor) 7 = Motor power (0 - PnMotor) 8 = Motor voltage (0 - UnMotor) 9 = DC link voltage (0-1000 V) 10 = PID Setpoint (0-100%) 11 = PID Feedback (0-100%) 12 = PID1 output (0-100%) 13 = Ext.PID output (0-100%)

Index	Parameter	Min	Max	Unit	Default	ID	Description
							14 = ProcessDataIn1 (0–100%) 15 = ProcessDataIn2 (0–100%) 16 = ProcessDataIn3 (0–100%) 17 = ProcessDataIn4 (0–100%) 18 = ProcessDataIn5 (0–100%) 19 = ProcessDataIn6 (0–100%) 20 = ProcessDataIn7 (0–100%) 21 = ProcessDataIn8 (0–100%) 22 = Block Out.1 (0–100%) 23 = Block Out.2 (0–100%) 24 = Block Out.3 (0–100%) 25 = Block Out.4 (0–100%) 26 = Block Out.5 (0–100%) 27 = Block Out.6 (0–100%) 28 = Block Out.7 (0–100%) 29 = Block Out.8 (0–100%) 30 = Block Out.9 (0–100%) 31 = Block Out.10 (0–100%)
P3.5.4.1.2	AO1 Filter Time	0.0	300.0	s	1.0 ⁽¹⁾	10051	0 = No filtering
P3.5.4.1.3	AO1 minimum	0	1		0 ⁽¹⁾	10052	
P3.5.4.1.4	AO1 minimum scale	-214748.36	214748.36	Varies	0.0 ⁽¹⁾	10053	0 = 0 mA/0 V 1 = 4 mA/2 V
P3.5.4.1.5	AO1 maximum scale	-214748.36	214748.36	Varies	0.0 ⁽¹⁾	10054	

¹ The selection of the application with parameter P1.2 Application gives the default value. See the default values in [11.25 Default Values of Parameters in the Different Applications](#).

The Analog Outputs of the Expander Slots C, D, and E

N O T I C E

Shows only the parameters for the outputs on option boards in slots C, D, and E. Make the selections as in Basic AO1 Function (P3.5.4.1.1). This group or these parameters are not visible if there are no digital outputs in slots C, D, or E.

6.6 Group 3.6: Fieldbus Data

Table 53: Fieldbus Data Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.6.1	Fieldbus Data Out 1 Selection	0	35000		1	852	
P3.6.2	Fieldbus Data Out 2 Selection	0	35000		2	853	
P3.6.3	Fieldbus Data Out 3 Selection	0	35000		3	854	
P3.6.4	Fieldbus Data Out 4 Selection	0	35000		4	855	
P3.6.5	Fieldbus Data Out 5 Selection	0	35000		5	856	

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.6.6	Fieldbus Data Out 6 Selection	0	35000		6	857	
P3.6.7	Fieldbus Data Out 7 Selection	0	35000		7	858	
P3.6.8	Fieldbus Data Out 8 Selection	0	35000		37	859	
P3.6.9	Fieldbus Data Out 9 Selection	0	35000		0	890	
P3.6.10	Fieldbus Data Out 10 Selection	0	35000		0	891	
P3.6.11	Fieldbus Data Out 11 Selection	0	35000		0	892	
P3.6.12	Fieldbus Data Out 12 Selection	0	35000		0	893	
P3.6.13	Fieldbus Data Out 13 Selection	0	35000		0	894	
P3.6.14	Fieldbus Data Out 14 Selection	0	35000		0	895	
P3.6.15	Fieldbus Data Out 15 Selection	0	35000		0	896	
P3.6.16	Fieldbus Data Out 16 Selection	0	35000		0	897	
P3.6.17	FB Sign of Actual	0	2		0	888	0 = Positive 1 = Actual direction 2 = Follow reference

Table 54: The Default Values for Process Data Out in Fieldbus

Data	Default Value	Scale
Process Data Out 1	Output frequency	0.01 Hz
Process Data Out 2	Motor speed	1 RPM
Process Data Out 3	Motor current	0.1 A
Process Data Out 4	Motor torque	0.1%
Process Data Out 5	Motor power	0.1%
Process Data Out 6	Motor voltage	0.1 V
Process Data Out 7	DC link voltage	1 V
Process Data Out 8	Last active fault code	1

Example

The value 2500 for output frequency equals 25.00 Hz, because the scale is 0.01. All the monitoring values in the Monitor menu (chapter [5.1 Uses of the Monitor Menu](#)) are given the scale value.

6.7 Group 3.7: Prohibit Frequencies

Table 55: Prohibit Frequencies Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.7.1	Prohibit Frequency Range 1 Low Limit	-1.00	320.00	Hz	0.00	509	0 = Not used
P3.7.2	Prohibit Frequency Range 1 High Limit	0.00	320.00	Hz	0.00	510	0 = Not used
P3.7.3	Prohibit Frequency Range 2 Low Limit	0.00	320.00	Hz	0.00	511	0 = Not used
P3.7.4	Prohibit Frequency Range 2 High Limit	0.00	320.00	Hz	0.00	512	0 = Not used
P3.7.5	Prohibit Frequency Range 3 Low Limit	0.00	320.00	Hz	0.00	513	0 = Not used
P3.7.6	Prohibit Frequency Range 3 High Limit	0.00	320.00	Hz	0.00	514	0 = Not used
P3.7.7	Ramp Time Factor	0.1	10.0	Times	1.0	518	

6.8 Group 3.8 Supervisions

Table 56: Supervision Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.8.1	Supervision #1 Item Selection	0	17		0	1431	0 = Output frequency 1 = Frequency reference 2 = Motor current 3 = Motor torque 4 = Motor power 5 = DC link voltage 6 = Analog input 1 7 = Analog input 2 8 = Analog input 3 9 = Analog input 4 10 = Analog input 5 11 = Analog input 6 12 = Temperature input 1 13 = Temperature input 2 14 = Temperature input 3 15 = Temperature input 4 16 = Temperature input 5 17 = Temperature input 6
P3.8.2	Supervision #1 Mode	0	2		0	1432	0 = Not used 1 = Low limit supervision 2 = High limit supervision
P3.8.3	Supervision #1 Limit	-50.00	50.00	Varies	25.00	1433	
P3.8.4	Supervision #1 Limit Hysteresis	0.00	50.00	Varies	5.00	1434	
P3.8.5	Supervision #2 Item Selection	0	17		1	1435	See P3.8.1

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.8.6	Supervision #2 Mode	0	2		0	1436	See P3.8.2
P3.8.7	Supervision #2 Limit	-50.00	50.00	Varies	40.00	1437	
P3.8.8	Supervision #2 Limit Hysteresis	0.00	50.00	Varies	5.00	1438	

6.9 Group 3.9: Protections

Table 57: General Protection Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.9.1.2	Response to External Fault	0	3		2	701	0 = No action 1 = Alarm 2 = Fault (Stop according to stop function) 3 = Fault (Stop by coasting)
P3.9.1.3	Input Phase Fault	0	1		0	730	0 = 3-phase support 1 = 1-phase support
P3.9.1.4	Undervoltage Fault	0	1		0	727	0 = Fault stored in history 1 = Fault not stored in history
P3.9.1.5	Response to Output Phase Fault	0	3		2	702	See P3.9.1.2.
P3.9.1.6	Response to Fieldbus Communication Fault	0	5		4	733	0 = No action 1 = Alarm 2 = Alarm + preset fault frequency (P3.9.1.13) 3 = Fault (Stop according to stop function) 4 = Fault (Stop by coasting)
P3.9.1.7	Slot Communication Fault	0	3		2	734	See P3.9.1.2.
P3.9.1.8	Thermistor Fault	0	3		0	732	See P3.9.1.2.
P3.9.1.9	PID Soft Fill Fault	0	3		2	748	See P3.9.1.2.
P3.9.1.10	Response to PID Supervision Fault	0	3		2	749	See P3.9.1.2.
P3.9.1.11	Response to External PID Supervision Fault	0	3		2	757	0 = No action 1 = Alarm 2 = Fault 3 = Fault, Coast
P3.9.1.12	Earth Fault	0	3		3	703	This parameter is available only in enclosure sizes MR7, MR8, MR9, MR10, MR11, and MR12, and also MR6 600 V and 690 V.
P3.9.1.13	Preset Alarm Frequency	P3.3.1.1	P3.3.1.2	Hz	25.00	183	

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.9.1.14	Response to Safe Torque Off (STO) Fault	0	2		2	775	0 = No action 1 = Alarm 2 = Fault (Stop by coasting)
P3.9.1.17	Fieldbus Communication Fault Force Delay	0.00	320.00	s	0.00	751	
P3.9.1.18	Fieldbus Communication Fault Clear Delay	0.00	320.00	s	0.00	752	
P3.9.1.19	Unbalance Current Alarm	0	1		1	753	

Table 58: Motor Thermal Protection Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.9.2.1	Motor Thermal Protection	0	3		2	704	0 = No action 1 = Alarm 2 = Fault (Stop by stop mode) 3 = Fault (Stop by coasting)
P3.9.2.2	Ambient Temperature	-20.0	100.0	°C	40.0	705	
P3.9.2.3	Zero Speed Cooling Factor	5.0	150.0	%	Varies	706	
P3.9.2.4	Motor Thermal Time Constant	1	200	min	Varies	707	
P3.9.2.5	Motor Thermal Loadability	10	150	%	100	708	

Table 59: Motor Stall Protection Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.9.3.1	Motor Stall Fault	0	3		0	709	0 = No action 1 = Alarm 2 = Fault (Stop according to stop mode) 3 = Fault (Stop by coasting)
P3.9.3.2	Stall Current	0.00	Is	A	Varies	710	
P3.9.3.3	Stall Time Limit	1.00	120.00	s	15.00	711	
P3.9.3.4	Stall Frequency Limit	1.00	P3.3.1.2	Hz	25.00	712	

Table 60: Motor Underload Protection Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.9.4.1	Underload Fault	0	3		0	713	0 = No action 1 = Alarm 2 = Fault (Stop according to stop mode)

Index	Parameter	Min	Max	Unit	Default	ID	Description
							3 = Fault (Stop by coasting)
P3.9.4.2	Underload Protection: Field Weakening Area Load	10.0	150.0	%	50.0	714	
P3.9.4.3	Underload Protection: Zero Frequency Load	5.0	150.0	%	10.0	715	
P3.9.4.4	Underload Protection: Time Limit	2.00	200.00	s	20.00	716	

Table 61: Quick Stop Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.9.5.1	Quick Stop Mode	0	2		Varies	1276	0 = Coasting 1 = Quick stop deceleration time 2 = Stop according to Stop function (P3.2.5)
P3.9.5.2	Quick Stop Activation	Varies	Varies		DigIN Slot0.2	1213	OPEN = Activated
P3.9.5.3	Quick Stop Deceleration Time	0.1	300.0	s	Varies	1256	
P3.9.5.4	Response to Quick Stop Fault	0	2		Varies	744	0 = No action 1 = Alarm 2 = Fault (Stop according to Quick stop mode)

Table 62: Temperature Input Fault 1 Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.9.6.1 ⁽¹⁾	Temperature Signal 1	0	63		0	739	B0 = Temperature Signal 1 B1 = Temperature Signal 2 B2 = Temperature Signal 3 B3 = Temperature Signal 4 B4 = Temperature Signal 5 B5 = Temperature Signal 6
P3.9.6.2 ⁽¹⁾	Alarm Limit 1	-30.0	200.0	°C	130.0	741	
P3.9.6.3 ⁽¹⁾	Fault Limit 1	-30.0	200.0	°C	155.0	742	
P3.9.6.4 ⁽¹⁾	Fault Limit Response 1	0	3		2	740	0 = No response 1 = Alarm 2 = Fault (Stop according to stop mode) 3 = Fault (Stop by coasting)

¹ Temperature input settings are only available if a B8 or BH option board is installed.

Table 63: Temperature Input Fault 2 Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.9.6.5 ⁽¹⁾	Temperature Signal 2	0	63		0	763	B0 = Temperature Signal 1 B1 = Temperature Signal 2 B2 = Temperature Signal 3 B3 = Temperature Signal 4 B4 = Temperature Signal 5 B5 = Temperature Signal 6
P3.9.6.6 ⁽¹⁾	Alarm Limit 2	-30.0	200.0	°C	130.0	764	
P3.9.6.7 ⁽¹⁾	Fault Limit 2	-30.0	200.0	°C	155.0	765	
P3.9.6.8 ⁽¹⁾	Fault Limit Response 2	0	3		2	766	0 = No response 1 = Alarm 2 = Fault (Stop according to stop mode) 3 = Fault (Stop by coasting)

¹ Temperature input settings are only available if a B8 or BH option board is installed.

Table 64: AI Low Protection Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.9.8.1	Analog Input Low Protection	0	2			767	0 = No protection 1 = Protection enabled in Run state 2 = Protection enabled in Run and Stop state
P3.9.8.2	Analog Input Low Fault	0	5		0	700	0 = No action 1 = Alarm 2 = Alarm + preset fault frequency (P3.9.1.13) 3 = Alarm + previous frequency reference 4 = Fault (Stop according to stop mode) 5 = Fault (Stop by coasting)

6.10 Group 3.10: Automatic Reset

Table 65: Automatic Reset Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.10.1	Automatic Reset	0	1		0 ⁽¹⁾	731	0 = Disabled 1 = Enabled
P3.10.2	Restart Function	0	1		1	719	0 = Flying start 1 = According to P3.2.4.
P3.10.3	Wait Time	0.10	10000.00	s	0.50	717	
P3.10.4	Trial Time	0.00	10000.00	s	60.00	718	
P3.10.5	Number of Trials	1	10		4	759	

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.10.6	Autoreset: Undervoltage	0	1		1	720	0 = No 1 = Yes
P3.10.7	Autoreset: Overvoltage	0	1		1	721	0 = No 1 = Yes
P3.10.8	Autoreset: Overcurrent	0	1		1	722	0 = No 1 = Yes
P3.10.9	Autoreset: AI Low	0	1		1	723	0 = No 1 = Yes
P3.10.10	Autoreset: Unit Overtemperature	0	1		1	724	0 = No 1 = Yes
P3.10.11	Autoreset: Motor Overtemperature	0	1		1	725	0 = No 1 = Yes
P3.10.12	Autoreset: External Fault	0	1		0	726	0 = No 1 = Yes
P3.10.13	Autoreset: Underload Fault	0	1		0	738	0 = No 1 = Yes
P3.10.14	Autoreset: PID Supervision Fault	0	1		0	776	0 = No 1 = Yes
P3.10.15	Autoreset: Ext PID Supervision Fault	0	1		0	777	0 = No 1 = Yes
P3.10.16	Fault Simulation	0	65535		0	1569	

¹ The selection of the application with parameter P1.2 Application gives the default value. See [11.25 Default Values of Parameters in the Different Applications](#).

6.11 Group 3.11: Application Settings

Table 66: Application Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.11.1	Password	0	9999		0	1806	
P3.11.2	C/F Selection	0	1		0 ⁽¹⁾	1197	0 = Celsius 1 = Fahrenheit
P3.11.3	kW/hp Selection	0	1		0 ⁽¹⁾	1198	0 = kW 1 = hp
P3.11.4	Multimonitor View	0	2		1	1196	0 = 2x2 sections 1 = 3x2 sections 2 = 3x3 sections

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.11.5	FUNCT Button Configuration	0	15		15	1195	B0 = Local/Remote B1 = Control Page B2 = Change Direction B3 = Quick Edit
P3.11.6	Set ParamLock	0	9999		0	1807	
P3.11.7	ParamLock Key	0	9999		0	1808	

¹ The default value in the US is 1.

6.12 Group 3.12: Timer Functions

Table 67: Interval 1

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.12.1.1	ON Time	00:00:00	23:59:59	hh:mm:ss	00:00:00	1464	
P3.12.1.2	OFF Time	00:00:00	23:59:59	hh:mm:ss	00:00:00	1465	
P3.12.1.3	Days					1466	B0 = Sunday B1 = Monday B2 = Tuesday B3 = Wednesday B4 = Thursday B5 = Friday B6 = Saturday
P3.12.1.4	Assign to Channel					1468	B0 = Time channel 1 B1 = Time channel 2 B2 = Time channel 3

Table 68: Interval 2

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.12.2.1	ON Time	00:00:00	23:59:59	hh:mm:ss	00:00:00	1469	See Interval 1.
P3.12.2.2	OFF Time	00:00:00	23:59:59	hh:mm:ss	00:00:00	1470	See Interval 1.
P3.12.2.3	Days					1471	See Interval 1.
P3.12.2.4	Assign to Channel					1473	See Interval 1.

Table 69: Interval 3

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.12.3.1	ON Time	00:00:00	23:59:59	hh:mm:ss	00:00:00	1474	See Interval 1.
P3.12.3.2	OFF Time	00:00:00	23:59:59	hh:mm:ss	00:00:00	1475	See Interval 1.
P3.12.3.3	Days					1476	See Interval 1.
P3.12.3.4	Assign to Channel					1478	See Interval 1.

Table 70: Interval 4

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.12.4.1	ON Time	00:00:00	23:59:59	hh:mm:ss	00:00:00	1479	See Interval 1.
P3.12.4.2	OFF Time	00:00:00	23:59:59	hh:mm:ss	00:00:00	1480	See Interval 1.
P3.12.4.3	Days					1481	See Interval 1.
P3.12.4.4	Assign to Channel					1483	See Interval 1.

Table 71: Interval 5

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.12.5.1	ON Time	00:00:00	23:59:59	hh:mm:ss	00:00:00	1484	See Interval 1.
P3.12.5.2	OFF Time	00:00:00	23:59:59	hh:mm:ss	00:00:00	1485	See Interval 1.
P3.12.5.3	Days					1486	See Interval 1.
P3.12.5.4	Assign to Channel					1488	See Interval 1.

Table 72: Timer 1

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.12.6.1	Duration	0	72000	s	0	1489	
P3.12.6.2	Timer 1				DigINSlot 0.1	447	
P3.12.6.3	Assign to Channel					1490	B0 = Time channel 1 B1 = Time channel 2 B2 = Time channel 3

Table 73: Timer 2

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.12.7.1	Duration	0	72000	s	0	1491	See Timer 1.
P3.12.7.2	Timer 2				DigINSlot 0.1	448	See Timer 1.
P3.12.7.3	Assign to Channel					1492	See Timer 1.

Table 74: Timer 3

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.12.8.1	Duration	0	72000	s	0	1493	See Timer 1.
P3.12.8.2	Timer 3				DigINSlot 0.1	449	See Timer 1.
P3.12.8.3	Assign to Channel					1494	See Timer 1.

6.13 Group 3.13: PID Controller

Table 75: PID Controller Basic Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.13.1.1	PID Gain	0.00	1000.00	%	100.00	118	
P3.13.1.2	PID Integration Time	0.00	600.00	s	1.00	119	
P3.13.1.3	PID Derivation Time	0.00	100.00	s	0.00	132	
P3.13.1.4	Process Unit Selection	1	46		1	1036	1 = % 2 = 1/min 3 = RPM 4 = ppm 5 = pps 6 = l/s 7 = l/min 8 = l/h 9 = kg/s 10 = kg/min 11 = kg/h 12 = m ³ /s 13 = m ³ /min 14 = m ³ /h 15 = m/s 16 = mbar 17 = bar 18 = Pa 19 = kPa 20 = mVS 21 = kW

Index	Parameter	Min	Max	Unit	Default	ID	Description
							22 = °C 23 = gal/s 24 = gal/min 25 = gal/h 26 = lb/s 27 = lb/min 28 = lb/h 29 = ft ³ /s 30 = ft ³ /min 31 = ft ³ /h 32 = ft/s 33 = in wg 34 = ft wg 35 = SPI 36 = lb/in ² 37 = psig 38 = hp 39 = °F 40 = ft 41 = inch 42 = mm 43 = cm 44 = m 45 = gpm 46 = cfm
P3.13.1.5	Process Unit Min	Varies	Varies	Varies	0	1033	
P3.13.1.6	Process Unit Max	Varies	Varies	Varies	100	1034	
P3.13.1.7	Process Unit Decimals	0	4		2	1035	
P3.13.1.8	Error Inversion	0	1		0	340	0 = Normal (Feedback < Setpoint -> Increase PID output) 1 = Inverted (Feedback < Setpoint -> Decrease PID output)
P3.13.1.9	Dead Band	0.00	99999.9 9	Varies	0	1056	
P3.13.1.10	Dead Band Delay	0.00	320.00	s	0.00	1057	

Table 76: Setpoint Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.13.2.1	Keypad setpoint 1	P3.13.1.5	P3.13.1.6	P3.13.1.4	0	167	
P3.13.2.2	Keypad setpoint 2	P3.13.1.5	P3.13.1.6	P3.13.1.4	0	168	

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.13.2.3	Setpoint ramp time	0.00	300.0	s	0.00	1068	
P3.13.2.4	PID setpoint boost activation	Varies	Varies		DigIN Slot0.1	1046	OPEN = No boost CLOSED = Boost
P3.13.2.5	PID select setpoint	Varies	Varies		DigIN Slot0.1 ⁽¹⁾	1047	OPEN = Setpoint 1 CLOSED = Setpoint 2
P3.13.2.6	Setpoint source 1 selection	0	32		3 ⁽¹⁾	332	0 = Not used 1 = Keypad setpoint 1 2 = Keypad setpoint 2 3 = AI1 4 = AI2 5 = AI3 6 = AI4 7 = AI5 8 = AI6 9 = ProcessDataIn1 10 = ProcessDataIn2 11 = ProcessDataIn3 12 = ProcessDataIn4 13 = ProcessDataIn5 14 = ProcessDataIn6 15 = ProcessDataIn7 16 = ProcessDataIn8 17 = Temperature input 1 18 = Temperature input 2 19 = Temperature input 3 20 = Temperature input 4 21 = Temperature input 5 22 = Temperature input 6 23 = Block Out.1 24 = Block Out.2 25 = Block Out.3 26 = Block Out.4 27 = Block Out.5 28 = Block Out.6 29 = Block Out.7 30 = Block Out.8 31 = Block Out.9 32 = Block Out.10
P3.13.2.7	Setpoint 1 minimum	-200.00	200.00	%	0.00	1069	
P3.13.2.8	Setpoint 1 maximum	-200.00	200.00	%	100.00	1070	
P3.13.2.9	Setpoint 1 boost	-2.0	2.0	x	1.0	1071	
P3.13.2.10	Setpoint source 2 selection	0	Varies		2 ⁽¹⁾	431	See P3.13.2.6.

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.13.2.11	Setpoint 2 minimum	-200.00	200.00	%	0.00	1073	See P3.13.2.7.
P3.13.2.12	Setpoint 2 maximum	-200.00	200.00	%	100.00	1074	See P3.13.2.8.
P3.13.2.13	Setpoint 2 boost	-2.0	2.0	x	1.0	1078	See P3.13.2.9.

¹ The selection of the application with parameter P1.2 Application gives the default value. See the default values in [11.25 Default Values of Parameters in the Different Applications](#).

Table 77: Feedback Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.13.3.1	Feedback Function	1	9		1 ⁽¹⁾	333	1 = Only Source1 in use 2 = SQRT(Source1); (Flow=Constant x SQRT(Pressure)) 3 = SQRT(Source1- Source 2) 4 = SQRT(Source 1) + SQRT (Source 2) 5 = Source 1 + Source 2 6 = Source 1 - Source 2 7 = MIN (Source 1, Source 2) 8 = MAX (Source 1, Source 2) 9 = MEAN (Source 1, Source 2)
P3.13.3.2	Feedback Function Gain	-1000.0	2100.0	%	100.0	1058	
P3.13.3.3	Feedback 1 Source Selection	0	30		2 ⁽¹⁾	334	0 = Not used 1 = AI1 2 = AI2 3 = AI3 4 = AI4 5 = AI5 6 = AI6 7 = ProcessDataIn1 8 = ProcessDataIn2 9 = ProcessDataIn3 10 = ProcessDataIn4 11 = ProcessDataIn5 12 = ProcessDataIn6 13 = ProcessDataIn7 14 = ProcessDataIn8 15 = Temperature input 1 16 = Temperature input 2 17 = Temperature input 3 18 = Temperature input 4 19 = Temperature input 5 20 = Temperature input 6 21 = Block Out.1 22 = Block Out.2 23 = Block Out.3 24 = Block Out.4 25 = Block Out.5 26 = Block Out.6 27 = Block Out.7

Index	Parameter	Min	Max	Unit	Default	ID	Description
							28 = Block Out.8 29 = Block Out.9 30 = Block Out.10
P3.13.3.4	Feedback 1 Minimum	-200.00	200.00	%	0.00	336	
P3.13.3.5	Feedback 1 Maximum	-200.00	200.00	%	100.00	337	
P3.13.3.6	Feedback 2 Source Selection	0	20		0	335	See P3.13.3.3.
P3.13.3.7	Feedback 2 Minimum	-200.00	200.00	%	0.00	338	See P3.13.3.4.
P3.13.3.8	Feedback 2 Maximum	-200.00	200.00	%	100.00	339	See P3.13.3.5.

¹ The selection of the application with parameter P1.2 Application gives the default value. See the default values in [11.25 Default Values of Parameters in the Different Applications](#).

Table 78: Feedforward Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.13.4.1	Feedforward Function	1	9		1	1059	See P3.13.3.1.
P3.13.4.2	Feedforward Function Gain	-1000	1000	%	100.0	1060	See P3.13.3.2.
P3.13.4.3	Feedforward 1 Source Selection	0	30		0	1061	See P3.13.3.3.
P3.13.4.4	Feedforward 1 Minimum	-200.00	200.00	%	0.00	1062	See P3.13.3.4.
P3.13.4.5	Feedforward 1 Maximum	-200.00	200.00	%	100.00	1063	See P3.13.3.5.
P3.13.4.6	Feedforward 2 Source Selection	0	30		0	1064	See P3.13.3.6.
P3.13.4.7	Feedforward 2 Min	-200.00	200.00	%	0.00	1065	See P3.13.3.7.
P3.13.4.8	Feedforward 2 Max	-200.00	200.00	%	100.00	1066	See P3.13.3.8.

Table 79: Sleep Function Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.13.5.1	SP1 Sleep Frequency	0.00	320.00	Hz	0.00	1016	0 = Not used
P3.13.5.2	SP1 Sleep Delay	0	3000	s	0	1017	0 = Not used
P3.13.5.3	SP1 Wake-up Level	-214748.36	214748.36	Varies	0.0000	1018	0 = Not used
P3.13.5.4	SP1 Wake-up Mode	0	1		0	1019	0 = Absolute level 1 = Relative setpoint

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.13.5.5	SP1 Sleep Boost	-99999.9 9	99999.9 9	P3.13.1.4	0	1793	
P3.13.5.6	SP1 Sleep Boost Maximum Time	1	300	s	30	1795	
P3.13.5.7	SP2 Sleep Frequency	0.00	320.00	Hz	0.00	1075	See P3.13.5.1.
P3.13.5.8	SP2 Sleep Delay	0	3000	s	0	1076	See P3.13.5.2.
P3.13.5.9	SP2 Wake-up Level	-214748. 36	214748. 36	Varies	0.0000	1077	See P3.13.5.3.
P3.13.5.10	SP2 Wake-up Mode	0	1		0	1020	0=Absolute Level 1=Relative Setpoint
P3.13.5.11	SP2 Sleep Boost	-99999.9 9	99999.9 9	P3.13.1.4	0	1794	See P3.13.5.5
P3.13.5.12	SP2 Sleep Boost Maximum Time	1	300	s	30	1796	See P3.13.5.6

Table 80: Feedback Supervision Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.13.6.1	Enable Feedback Supervision	0	1		0	735	0 = Disabled 1 = Enabled
P3.13.6.2	Upper Limit	-99999.9 9	99999.9 9	Varies	Varies	736	
P3.13.6.3	Lower Limit	-99999.9 9	99999.9 9	Varies	Varies	758	
P3.13.6.4	Delay	0	30000	s	0	737	
P3.13.6.5	Response to PID Supervision Fault	0	3		2	749	0 = No action 1 = Alarm 2 = Fault (Stop according to stop mode) 3 = Fault (Stop by coasting)

Table 81: Pressure Loss Compensation Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.13.7.1	Enable Setpoint 1	0	1		0	1189	0 = Disabled 1 = Enabled
P3.13.7.2	Setpoint 1 Max Compensation	-99999.9 9	99999.9 9	Varies	Varies	1190	
P3.13.7.3	Enable Setpoint 2	0	1		0	1191	See P3.13.7.1.
P3.13.7.4	Setpoint 2 Max Compensation	-99999.9 9	99999.9 9	Varies	Varies	1192	See P3.13.7.2.

Table 82: Soft Fill Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.13.8.1	Soft Fill Function	0	2		0	1094	0 = Disabled 1 = Enabled, Level 2 = Enabled, Timeout
P3.13.8.2	Soft Fill Frequency	0.00	P3.3.1.2	Hz	0	1055	
P3.13.8.3	Soft Fill Level	-99999.9 9	99999.9 9	Varies	0.0000	1095	
P3.13.8.4	Soft Fill Timeout	0	30000	s	0	1096	0 = No timeout, no fault triggering
P3.13.8.5	D Soft Fill Fault	0	3		2	748	0 = No action 1 = Alarm 2 = Fault (Stop according to stop mode) 3 = Fault (Stop by coasting)

Table 83: Input Pressure Supervision Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.13.9.1	Enable Supervision	0	1		0	1685	0 = Disabled 1 = Enabled
P3.13.9.2	Supervision Signal	0	23		0	1686	0 = Analog input 1 1 = Analog input 2 2 = Analog input 3 3 = Analog input 4 4 = Analog input 5 5 = Analog input 6 6 = ProcessDataIn1 (0–100%) 7 = ProcessDataIn2 (0–100%) 8 = ProcessDataIn3 (0–100%) 9 = ProcessDataIn4 (0–100%) 10 = ProcessDataIn5 (0–100%) 11 = ProcessDataIn6 (0–100%) 12 = ProcessDataIn7 (0–100%) 13 = ProcessDataIn8 (0–100%) 14 = Block Out.1 15 = Block Out.2 16 = Block Out.3 17 = Block Out.4 18 = Block Out.5 19 = Block Out.6 20 = Block Out.7 21 = Block Out.8 22 = Block Out.9 23 = Block Out.10
P3.13.9.3	Supervision Unit Selection	0	9	Varies	3	1687	1 = % 2 = mbar 3 = bar

Index	Parameter	Min	Max	Unit	Default	ID	Description
							4 = Pa 5 = kPa 6 = PSI 7 = mmHg 8 = Torr 9 = lb/in ²
P3.13.9.4	Supervision Unit Decimals	0	4		2	1688	
P3.13.9.5	Supervision Unit Minimum Value	-99999.99	99999.99	P3.13.9.3	0.00	1689	
P3.13.9.6	Supervision Unit Maximum Value	-99999.99	99999.99	P3.13.9.3	10.0	1690	
P3.13.9.7	Supervision Alarm Level	P3.13.9.5	P3.13.9.6	P3.13.9.3	Varies	1691	
P3.13.9.8	Supervision Fault Level	P3.13.9.5	P3.13.9.6	P3.13.9.3	0.10	1692	
P3.13.9.9	Supervision Fault Delay	0.00	60.00	s	5.00	1693	
P3.13.9.10	PID Setpoint Reduction	0.0	100.0	%	10.0	1694	
V3.13.9.11	Input Pressure	P3.13.9.5	P3.13.9.6	P3.13.9.3	Varies	1695	This monitoring value shows the actual value of the pump input pressure.

Table 84: Sleep - No Demand Detected

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.13.10.1	Sleep NoDemand Detection Enable	0	1		0	1649	0 = No 1 = Yes
P3.13.10.2	SNDD Error Hysteresis	0	99999.9	P3.13.1.4	0.5	1658	
P3.13.10.3	SNDD FrequencyHysteresis	0.00	P3.3.1.2	Hz	3.00	1663	
P3.13.10.4	SNDD Supervision Time	0	600	s	120	1668	
P3.13.10.5	SNDD Actual Add	0.00	P3.13.10.2	P3.13.1.4	0.5	1669	

Table 85: Multi-setpoint Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.13.12.1	Multi-Setpoint0	P3.13.1.5	P3.13.1.6	P3.13.1.4	0.0	15560	
P3.13.12.2	Multi-Setpoint1	P3.13.1.5	P3.13.1.6	P3.13.1.4	0.0	15561	
P3.13.12.3	Multi-Setpoint2	P3.13.1.5	P3.13.1.6	P3.13.1.4	0.0	15562	
P3.13.12.4	Multi-Setpoint3	P3.13.1.5	P3.13.1.6	P3.13.1.4	0.0	15563	

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.13.12.5	Multi-Setpoint4	P3.13.1.5	P3.13.1.6	P3.13.1.4	0.0	15564	
P3.13.12.6	Multi-Setpoint5	P3.13.1.5	P3.13.1.6	P3.13.1.4	0.0	15565	
P3.13.12.7	Multi-Setpoint6	P3.13.1.5	P3.13.1.6	P3.13.1.4	0.0	15566	
P3.13.12.8	Multi-Setpoint7	P3.13.1.5	P3.13.1.6	P3.13.1.4	0.0	15567	
P3.13.12.9	Multi-Setpoint8	P3.13.1.5	P3.13.1.6	P3.13.1.4	0.0	15568	
P3.13.12.10	Multi-Setpoint9	P3.13.1.5	P3.13.1.6	P3.13.1.4	0.0	15569	
P3.13.12.11	Multi-Setpoint10	P3.13.1.5	P3.13.1.6	P3.13.1.4	0.0	15570	
P3.13.12.12	Multi-Setpoint11	P3.13.1.5	P3.13.1.6	P3.13.1.4	0.0	15571	
P3.13.12.13	Multi-Setpoint12	P3.13.1.5	P3.13.1.6	P3.13.1.4	0.0	15572	
P3.13.12.14	Multi-Setpoint13	P3.13.1.5	P3.13.1.6	P3.13.1.4	0.0	15573	
P3.13.12.15	Multi-Setpoint14	P3.13.1.5	P3.13.1.6	P3.13.1.4	0.0	15574	
P3.13.12.16	Multi-Setpoint15	P3.13.1.5	P3.13.1.6	P3.13.1.4	0.0	15575	
P3.13.12.17	Multi-SetpointSelection 0				DigINSlot0.1	15576	
P3.13.12.18	Multi-SetpointSelection 1				DigINSlot0.1	15577	

6.14 Group 3.14: External PID Controller

Table 86: Basic Settings for the External PID Controller

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.14.1.1	Enable External PID	0	1		0	1630	0 = Disabled 1 = Enabled
P3.14.1.2	Start Signal				DigIN Slot0.2	1049	OPEN = PID2 in stop mode CLOSED = PID2 regulating
P3.14.1.3	Output in Stop	0.0	100.0	%	0.0	1100	
P3.14.1.4	PID Gain	0.00	1000.00	%	100.00	1631	See P3.13.1.1
P3.14.1.5	PID Integration Time	0.00	600.00	s	1.00	1632	See P3.13.1.2
P3.14.1.6	PID Derivation Time	0.00	100.00	s	0.00	1633	See P3.13.1.3
P3.14.1.7	Process Unit Selection	0	37		0	1635	See P3.13.1.4
P3.14.1.8	Process Unit Min	Varies	Varies	Varies	0	1664	See P3.13.1.5
P3.14.1.9	Process Unit Max	Varies	Varies	Varies	100	1665	See P3.13.1.6
P3.14.1.10	Process Unit Decimals	0	4		2	1666	See P3.13.1.7

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.14.1.11	Error Inversion	0	1		0	1636	See P3.13.1.8
P3.14.1.12	Dead Band	Varies	Varies	Varies	0.0	1637	See P3.13.1.9
P3.14.1.13	Dead Band Delay	0.00	320.00	s	0.00	1638	See P3.13.1.10

Table 87: Setpoints of the External PID Controller

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.14.2.1	Keypad Setpoint 1	P3.14.1.8	P3.14.1.9	Varies	0.00	1640	
P3.14.2.2	Keypad Setpoint 2	P3.14.1.8	P3.14.1.9	Varies	0.00	1641	
P3.14.2.3	Setpoint Ramp Time	0.00	300.00	s	0.00	1642	
P3.14.2.4	Select Setpoint				DigIN Slot0.1	1048	OPEN = Setpoint 1 CLOSED = Setpoint 2
P3.14.2.5	Setpoint Source 1 Selection	0	32		1	1643	0 = Not Used 1 = Keypad Setpoint 1 2 = Keypad Setpoint 2 3 = AI1 4 = AI2 5 = AI3 6 = AI4 7 = AI5 8 = AI6 9 = ProcessDataIn1 10 = ProcessDataIn2 11 = ProcessDataIn3 12 = ProcessDataIn4 13 = ProcessDataIn5 14 = ProcessDataIn6 15 = ProcessDataIn7 16 = ProcessDataIn8 17 = Temperature Input 1 18 = Temperature Input 2 19 = Temperature Input 3 20 = Temperature Input 4 21 = Temperature Input 5 22 = Temperature Input 6 23 = Block Out.1 24 = Block Out.2 25 = Block Out.3 26 = Block Out.4 27 = Block Out.5 28 = Block Out.6 29 = Block Out.7

Index	Parameter	Min	Max	Unit	Default	ID	Description
							30 = Block Out.8 31 = Block Out.9 32 = Block Out.10 If temperature inputs are selected, the values of parameters P3.14.1.8 Process Unit Min and P3.14.1.9 Process Unit Max must be set to correspond to the scale of the temperature measurement board.
P3.14.2.6	Setpoint 1 minimum	-200.00	200.00	%	0.00	1644	
P3.14.2.7	Setpoint 1 maximum	-200.00	200.00	%	100.00	1645	
P3.14.2.8	Setpoint source 2 selection	0	32		2	1646	See P3.14.2.5.
P3.14.2.9	Setpoint 2 minimum	-200.00	200.00	%	0.00	1647	
P3.14.2.10	Setpoint 2 maximum	-200.00	200.00	%	100.00	1648	

Table 88: Feedback of the External PID Controller

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.14.3.1	Feedback Function	1	9		1	1650	See P3.13.3.1
P3.14.3.2	Feedback Function Gain	-1000.0	1000.0	%	100.0	1651	See P3.13.3.2
P3.14.3.3	Feedback 1 Source Selection	0	30		2	1652	See P3.13.3.3
P3.14.3.4	Feedback 1 Minimum	-200.00	200.00	%	0.00	1653	
P3.14.3.5	Feedback 1 Maximum	-200.00	200.00	%	100.00	1654	
P3.14.3.6	Feedback 2 Source Selection	0	30		0	1655	See P3.13.3.6.
P3.14.3.7	Feedback 2 Minimum	-200.00	200.00	%	0.00	1656	
P3.14.3.8	Feedback 2 Maximum	-200.00	200.00	%	100.00	1657	

Table 89: Process Supervision of the External PID Controller

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.14.4.1	Enable Supervision	0	1		0	1659	0 = Disabled 1 = Enabled
P3.14.4.2	Upper Limit	Varies	Varies	Varies	0	1660	See P3.13.6.2

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.14.4.3	Lower Limit	Varies	Varies	Varies	0	1661	See P3.13.6.3
P3.14.4.4	Delay	0	30000	s	0	1662	
P3.14.4.5	Response to External PID Supervision Fault	0	3		2	757	See P3.9.1.2

6.15 Group 3.15: Multi-pump

Table 90: Multi-pump Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.15.1	Multi-pump Mode	0	2		0 ⁽¹⁾	1785	0 = Single Drive 1 = Multifollower 2 = Multimaster
P3.15.2	Number of Pumps	1	8		1 ⁽¹⁾	1001	
P3.15.3	Pump ID Number	1	8		0	1500	
P3.15.4	Start and Feedback Signals	0	2		1	1782	0 = Not Connected 1 = Only Start Signal Connected 2 = Both Signals Connected
P3.15.5	Pump Interlocking	0	1		1 ⁽¹⁾	1032	0 = Not used 1 = Enabled
P3.15.6	Autochange	0	2		1 ⁽¹⁾	1027	0 = Disabled 1 = Enabled (interval) 2 = Enabled (weekdays)
P3.15.7	Autochanged Pumps	0	1		1 ⁽¹⁾	1028	0 = Auxiliary pumps 1 = All pumps
P3.15.8	Autochange Interval	0.0	3000.0	h	48.0 ⁽¹⁾	1029	
P3.15.9	Autochange Days	0	127		0	1786	B0 = Sunday B1 = Monday B2 = Tuesday B3 = Wednesday B4 = Thursday B5 = Friday B6 = Saturday
P3.15.10	Autochange: Time of Day	00:00:00	23:59:59	Time	00:00:00	1787	
P3.15.11	Autochange: Frequency Limit	0.00	P3.3.1.2	Hz	25.00 ⁽¹⁾	1031	
P3.15.12	Autochange: PumpLimit	0	8		1 ⁽¹⁾	1030	
P3.15.13	Bandwidth	0	100	%	10 ⁽¹⁾	1097	Setpoint = 5 bar Bandwidth = 10%.
P3.15.14	Bandwidth Delay	0	3600	s	10 ⁽¹⁾	1098	